

Hardware Information Web Interface (CGI)

Objective

Implement a simple **CGI-based web application** in qemu that reads qemu information from the Linux system (/sys filesystem) and displays it in a web browser.

Task

You are required to build a CGI program that:

◆ 1. Reads System Information

From the /sys filesystem (as explained in the lab), including:

- Network:
 - MAC Address
 - Interface Speed
 - RX Bytes
 - Storage:
 - Block device (mmcblk0)
 - Number of sectors
 - Logical block size
-

◆ 2. Compute Storage Size

Using the read values, calculate:

- Total size in **Bytes**
 - Total size in **MB**
 - Total size in **GB**
-

◆ 3. Display in Browser

- Output must be displayed using a **CGI program**
- The result should be formatted as a simple HTML page
- Accessible via browser:

`http://<board-ip>/cgi-bin/<program>`

e.g. `http://192.168.0.100/cgi-bin/<program>`

⚠ Important Requirement (Core of the Task)

While implementing the size calculation:

- You may notice that the **converted values (MB/GB) do not appear correctly**
 - You must:
 - Identify the reason
 - Fix the issue in your implementation
-

💡 Notes

- Required paths are already covered in the lab (e.g. `/sys/block/mmcblk0/...`)
-

📦 Submission

- The assignment will be checked **live during the lab**
 - Each student must run the solution on their own environment
-

📅 Deadline

Monday, 20 April (during lab session)